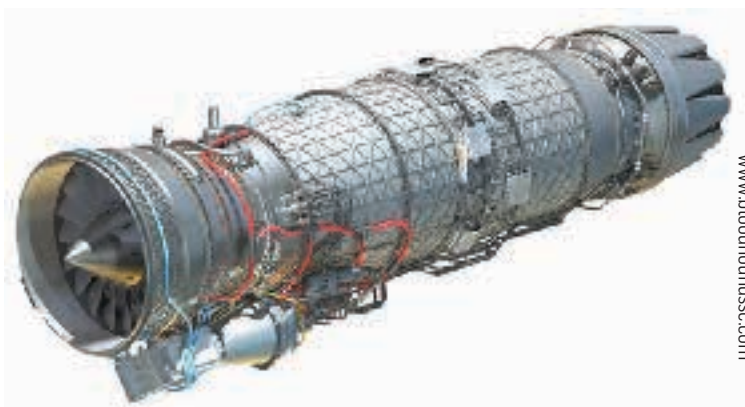


Jet Engine

We often take for granted how easily a plane weighing over half a million pounds lifts off the ground with such ease. This miracle can be accredited to jet engines – an invention that uses Newton's third law of physics to create 'thrust'. Developed in the 1930s by Hans von Ohain in Germany and Frank Whittle in Britain, the modern jet engine represents the 80-year evolution of the gas turbine – a kind of rotating device that extracts energy from a fluid flow and converts it into energy.

Jet engines are a form of 'air-breathing' gas turbines in which the fluid (air) is compressed, mixed with fuel and burned at high temperature and pressure to create a flow of hot gas that spins the turbine. This jet of gas then streams out the back, creating a huge amount of thrust – equivalent almost to the power of 1,500 cars, enabling the plane to take off into air despite its heavy weight.



What's interesting to note is that in the case of jet engines, bigger is not always better. To measure the power of a jet engine, its thrust-to-weight ratio is more important than its size. Fighter jets and Turbo jets used for the fast warplanes and the supersonic Concorde therefore had higher thrust-to-weight ratios to enable high-speed manoeuvres and near-vertical climbs. However, though fast, these jet engines consume bucket-loads of fuel and are therefore not appropriate for commercial airplanes that need to maintain cruising speed while using fuel efficiently. It's thanks to the jet engine that today we're able to travel distances at speeds that would have been impossible 80 years ago. This invention truly changed the face of aviation forever.